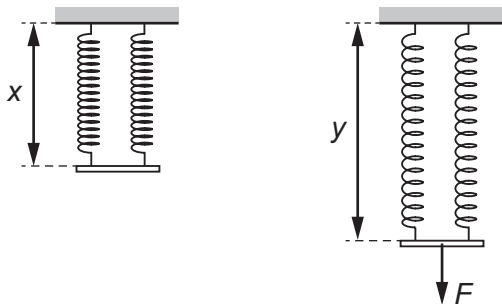


- 22** Two identical springs have the same spring constant  $k$ . The springs are connected in parallel. The length of the unstretched springs is  $x$ .

A force  $F$  is applied to the spring combination. The length of the springs is now  $y$ .



Both springs are deformed within their limits of proportionality.

Which expression gives the elastic potential energy stored in **one** of the springs?

- A**  $\frac{1}{4}F(y - x)$       **B**  $\frac{1}{2}F(y - x)$       **C**  $\frac{1}{4}k(y - x)^2$       **D**  $k(y - x)^2$